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PRACTICAL NO:13

AIM:- Mote-to-Mote Radio Communication Simulation

Simulate packet transmission between two motes and analyze signal strength and packet loss under varying radio gain values.

Mote.ned:

```
package motecommunication;

simple Mote
{
    parameters:
        double sendInterval @unit(s);
    gates:
        input in;
        output out;
}
```

RadioNetwork.ned:

```
package motecommunication;

network RadioNetwork
{
    submodules:
        mote1: Mote
        {
            parameters:
                sendInterval = 1s;
        }

        mote2: Mote
        {
            parameters:
                sendInterval = 0s;
        }

    connections:
        mote1.out --> mote2.in;
        mote2.out --> mote1.in;
}
```

Mote.cc:

```
#include <omnetpp.h>

class Mote : public cSimpleModule
{
private:
    cMessage *timer;
    int received;

protected:

    virtual void initialize()
    {
```

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```
received = 0;

if(strcmp(getName(),"mote1")==0)
{
    timer = new cMessage("sendTimer");
    scheduleAt(simTime()+1,timer);
}
else
    timer = NULL;
}

virtual void handleMessage(cMessage *msg)
{
    if(msg==timer)
    {
        EV << "Sending Packet\n";

        cMessage *pkt=new cMessage("DataPacket");

        send(pkt,"out");

        scheduleAt(simTime()+1,timer);
    }
    else
    {
        received++;

        EV << "Packet Received Count="
            << received << "\n";

        delete msg;
    }
}

virtual ~Mote()
{
    cancelAndDelete(timer);
}
};

Define_Module(Mote);
```

Omnetpp.ini:

```
[General]
network = motecommunication.RadioNetwork

sim-time-limit = 20s
```

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